

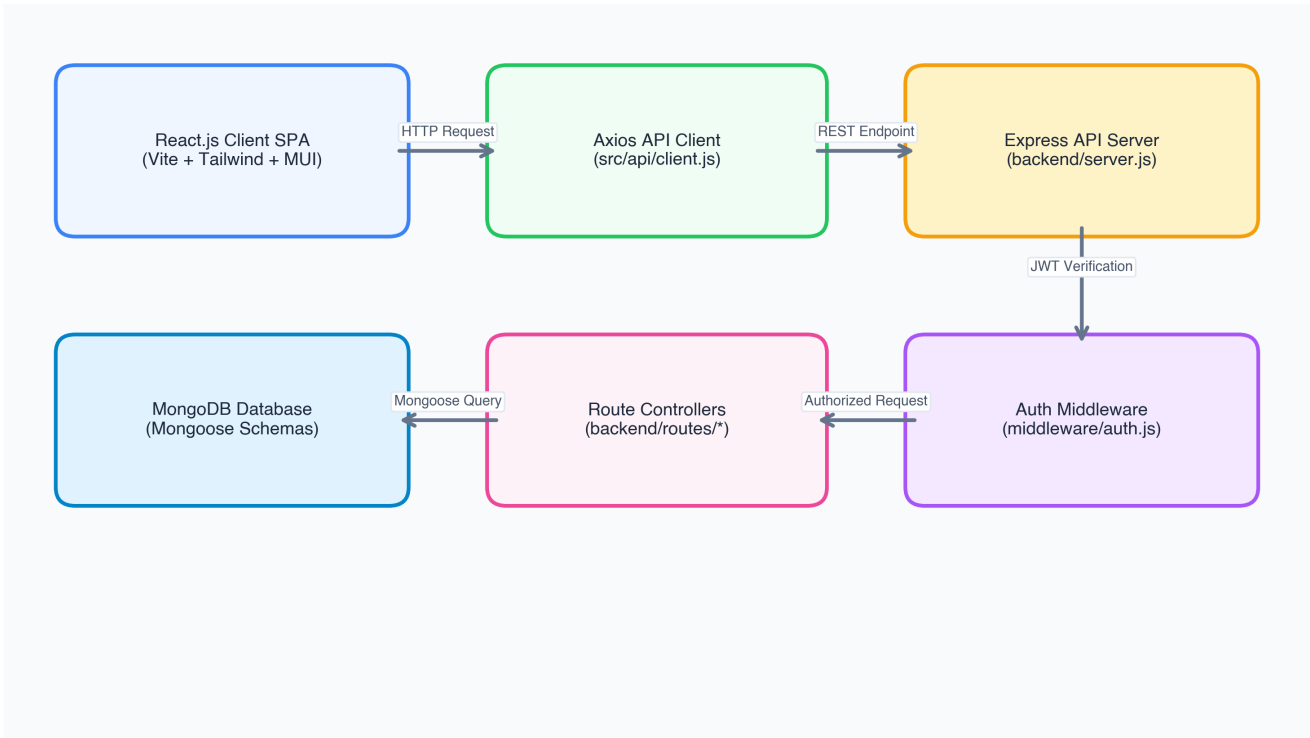
SpacECE Smart Parenting Portal

1. Executive Summary & Project Overview

The **SpacECE Smart Parenting & Early Childhood Education (ECE) Portal** is an enterprise-grade full-stack MERN application (MongoDB, Express.js, React.js, Node.js) bundled with Vite. Designed for early learners, parents, educators, and administrators, the system combines gamified learning worlds across **5 NEP 2020 developmental domains** (Literacy, Numeracy, Cognitive, Creativity, Emotional) with automated milestone tracking, diagnostic analytics, and personalized AI recommendations.

2. High-Level System Architecture & Flow

SpacECE Smart Parenting Portal — System Architecture & Data Flow



Architectural Flow: The client web application renders through React single-page routes secured by role-based route guards. API interaction is processed via an Axios HTTP client with JWT middleware authorization. Express.js API handlers execute business logic and persist records to MongoDB collections through strongly-typed Mongoose schemas.

3. Root Workspace Skeleton (√)

```
SpacECE MongoDB/  
■■■ backend/           # Node.js + Express.js REST API Server  
■■■ frontend/          # React.js + Vite Single Page Application (SPA)  
■■■ run.sh              # Dual-launcher shell script (starts Express & Vite)  
■■■ spacece_games_list.csv # Master game module index catalog  
■■■ PROJECT_STRUCTURE.md # Markdown documentation specification  
■■■ README.md           # Project quickstart guide & instructions
```

4. Backend Subsystem Architecture (`/backend`)

The backend is structured into modular layers separation: Express application server setup (server.js), Mongoose Data Models (models/), and REST API Endpoint Routers (routes/).

Database Schemas & Models (`backend/models/`)

Model File	Collection	Description & Key Fields
User.js	users	User authentication accounts (email, password hash, role: parent/teacher/admin, emailVerified).
ChildProfile.js	childprofiles	Child profile records (name, age, parent_uid, xp, stars, coins, level, mascot, module progress).
Score.js	scores	Activity performance logs (child_id, game_id, activity_type, score, accuracy, time_spent).
Literacy.js	literacies	Literacy game configurations for phonics, stories, word builder, and vocabulary.
CognitiveSel.js	cognitivesels	Brain world, emotion world, and SEL branching story configurations.
Recommendation.js	recommendations	AI & Educator focus activity recommendations (child_id, domain, reason, priority).
Notification.js	notifications	Alert & system notification messages (recipient_uid, title, body, read_status).
Center.js	centers	Early Childhood Learning Centers directory records.

API Routers (`backend/routes/`)

Route File	Base Path	Endpoints & Business Responsibilities
auth.js	/api/auth	User registration, password login, Google OAuth 2.0, password resets, email change.
children.js	/api/children	Child profile CRUD, mascot updates, awardProgress XP/stars/coins additions.
scores.js	/api/scores	Game score submissions, domain accuracy calculations, & session logs.
literacy.js	/api/literacy	Retrieval & configuration endpoints for Phonics, Stories, and Vocabulary games.
cognitiveSel.js	/api/cognitive-sel	Content endpoints for Emotion Check-In, Recognition, and Branching Stories.
milestones.js	/api/milestones	NEP 2020 milestone assessment mapping and recommendation generation.
notifications.js	/api/notifications	Parent notification feeds and unread status counters.

5. Frontend Subsystem Architecture (`/frontend`)

The frontend application is organized cleanly under frontend/src/ into specialized directories for API communication, global state contexts, page views, UI components, and router security.

Frontend Application Structure (`src/`)

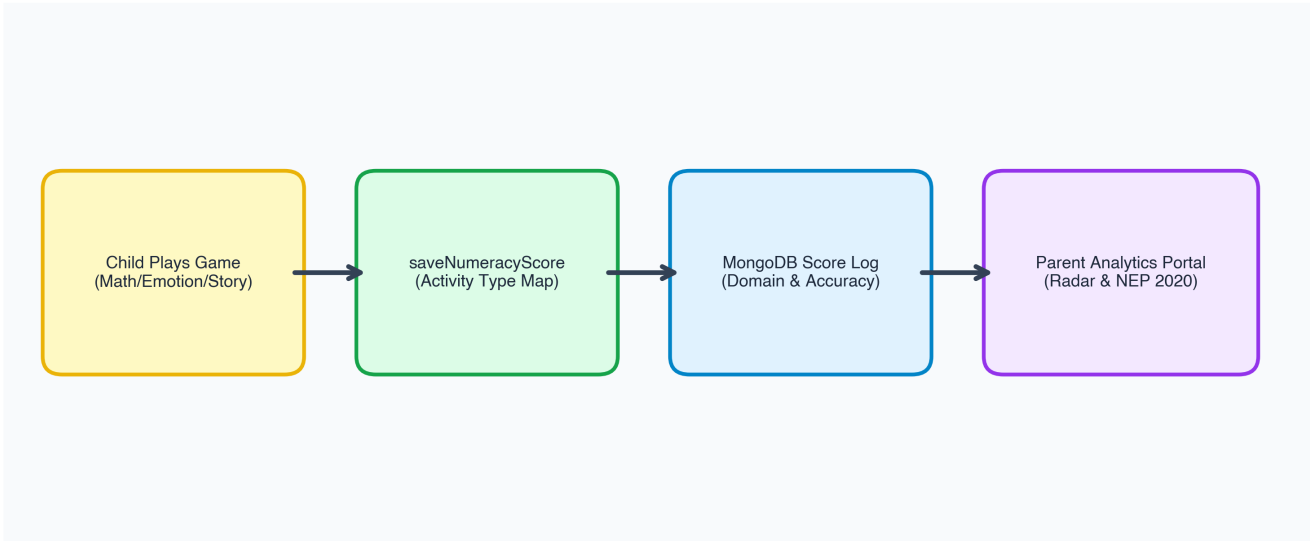
frontend/src/	
■■■■ api/	# Axios client & API service abstractions (authService, services, literacyService)
■■■■ components/	# UI components (animations, auth, child, layout, literacy, shared, theme)
■■■■ context/	# Context Providers (AuthContext, UserContext, ChildProfileContext, NotificationContext)
■■■■ pages/	# View pages (admin, analytics, auth, child, cognitive-sel, literacy, numeracy, parent, teacher)
■■■■ routes/	# Protected & Role-based route guards (ProtectedRoute.jsx, RoleRoute.jsx)
■■■■ services/	# Analytics calculation engine (analyticsService.ts)
■■■■ utils/	# Helpers & multilingual dictionary (helpers.js, translations.js)
■■■■ App.jsx	# Central React Router tree & application layout setup
■■■■ index.css	# Global design system tokens, themes, & animation keyframes
■■■■ main.jsx	# React application entry point & root DOM mounting

Page Modules Breakdown (`src/pages/`)

Category Folder	Primary Pages	Functionality & Learning Goals
auth/	Login, Register, RoleSelection, Terms, Privacy	Authentication flows with role pre-selection and Google OAuth 2.0 integration.
child/	Home (Dashboard), Adventure, AvatarPage, Awards	Kid-friendly home, 3D island map explorer, avatar customizer, and reward trophies.
numeracy/	MathWorld, PuzzleWorld, LogicIsland, NumberAdventure	Counting, shape sides, size comparison, number matching, arithmetic & mazes.
cognitive-sel/	EmotionWorldPage, StoryWorldPage, SpatialBrainPage	Emotion check-in, recognition, friendship stories, and branching narratives.
literacy/	ReadingWorldPage, PhonicsLandPage, StoryWorldPage	Phonics letter sounds, interactive story reader, sight words & picture matching.
analytics/	ParentDashboard, DevelopmentalLearningCurves	NEP 2020 domain competency radar chart, time allocation, & readiness scores.
parent/ & teacher/	ParentDashboard, TeacherDashboard, ChildrenList	Child profile management, classroom roster tracking, focus recommendations.

6. Score Logging & Analytics Pipeline

End-to-End Game Score Logging & Analytics Pipeline



Domain Mapping Pipeline: When a child completes a game round across Math World, Emotion World, or Phonics Land, `saveNumeracyScore` maps the `game_id` to its explicit developmental domain (*Literacy, Numeracy, Cognitive, Creativity, Emotional*). The score is logged to MongoDB via `/api/scores`. When the parent or educator views the Analytics Portal, `analyticsService.ts` aggregates scores by domain to compute the **School Readiness Score** and render the **NEP 2020 Competency Radar Chart**.

7. Role Matrix & Responsibilities

Role	Access Paths	Key Capabilities & Permissions
Child	/child/*, /math-world, /child/emotion-world	Play games, earn XP/stars/coins, update avatar, view learning roadmap.
Parent	/parent/*, /analytics/*, /settings/*	Manage child profiles, view NEP 2020 milestone reports, manage settings.
Teacher	/teacher/*, /analytics/teacher	View classroom student rosters, analyze learning gaps, assign focus activities.
Admin	/admin/*	System-wide metrics, user account controls, platform thresholds configuration.